ENT ATLAS

IMAGING

Imaging conditions: Nose, sinuses and orbit

Key facts and case images for the core conditions, then the advanced and rare

· November 2026

IMAGING

Core conditions

The common conditions every resident should recognise, each with its key imaging findings, top differentials, a short checklist and representative scans.

NOSE & NASAL CAVITY · CONGENITAL

Meningoencephalocele

IMAGING FINDINGS

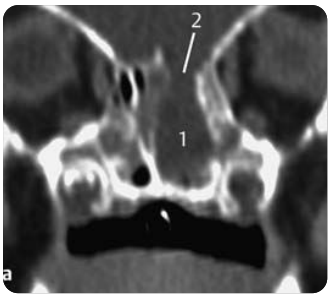
- Bony defect in the cribriform plate, lateral lamella, ethmoid roof or sphenoid
- Soft tissue or fluid through the defect
- MRI: CSF, meninges or brain in continuity with the intracranial contents

TOP DIFFERENTIALS

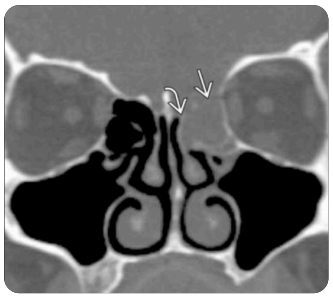
- Skull base defect without leak
- Sinus opacification from mucus

CHECKLIST

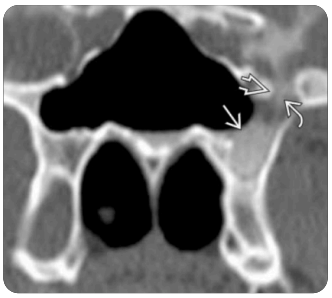
- Lateral sphenoid recess and multiple defects
- Signs of raised intracranial pressure: empty sella



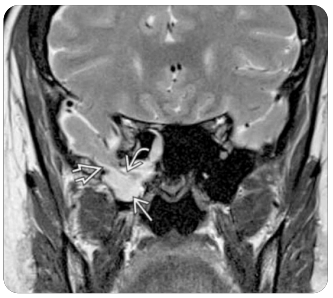
CT, coronal. The left nasal cavity is filled with a soft-tissue mass (1) that seems to be connected to brain through a defect in the roof of the…



Coronal bone CT shows large bony defect in left ethmoid roof, lateral to insertion of middle turbinate. Because there is complete opacification of the…



Coronal bone CT after intrathecal contrast shows CSF in left sphenoid chamber, bone defect, and contrast extending through defect. CT…



Coronal T2WI MR reveals lateral sphenoid sinus filled with CSF. Note the osseous defect in lateral sphenoid roof, and brain herniating through…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 920.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Maxillary Sinusitis

IMAGING FINDINGS

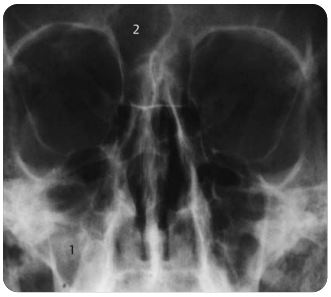
- Air–fluid level, frothy or bubbly secretions
- Mucosal thickening
- Chronic: sclerotic thickened walls (osteitis)

TOP DIFFERENTIALS

- Retention cyst or polyp
- Post-traumatic blood
- Odontogenic source (periapical lucency, oroantral fistula)

CHECKLIST

- Look at the teeth and the ostiomeatal unit
- Contrast CT only for complications



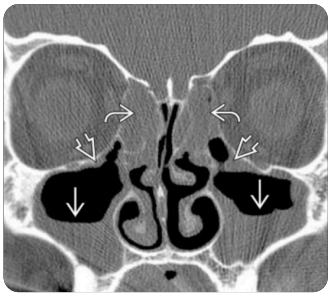
Caldwell view. This 26-year-old patient had pain in the right maxillary region. The right maxillary sinus seems to be opacified in the lower part (1). The lef…



Axial NECT in patient with purulent nasal discharge and headache shows patchy bilateral ethmoid sinus disease, an air-fluid level in the right sphenoid…



Coronal reconstruction NECT demonstrates bubbly, frothy secretions in sphenoid sinuses in a patient with clinical diagnosis of ABR…



Coronal bone CT shows acute bilateral maxillary and ethmoid sinusitis. Note airfluid levels and mucosal thickening in maxillary sinuses and complete…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 688.

NOSE & NASAL CAVITY · INFECTION & INFLAMMATION

Nasal Polyps

IMAGING FINDINGS

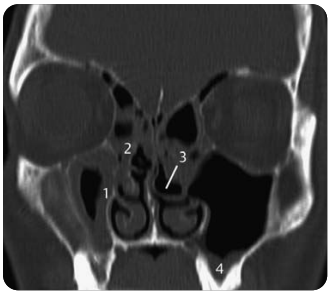
- Polypoid soft tissue filling the nose and all sinuses, both sides
- Widening of the ethmoid labyrinth and thinned septa
- Hyperdense contents suggest fungus or inspissated mucus

TOP DIFFERENTIALS

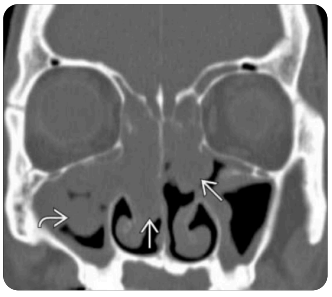
- Allergic fungal sinusitis
- Cystic fibrosis
- Inverted papilloma (unilateral)

CHECKLIST

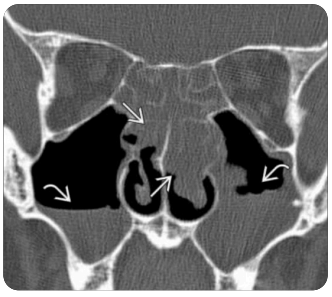
- Bone erosion or asymmetry: think tumour or fungus
- Pre-operative CLOSE checklist



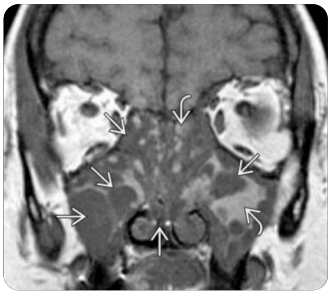
CT, coronal. The scan shows mucosal thickening in the right maxillary sinus (1), partial opacification of the ethmoid cells (2) on both sides, pneumatizati…



Coronal bone CT shows the classic appearance of sinonasal polyposis
Koch et al., Diagnostic Imaging: Head and Neck, 3rd ed., p. 708



Coronal bone CT in a patient with polyposis and new onset of acute sinusitis symptoms shows polyps in the nasal cavity occluding the middle…



Coronal T1WI MR shows multiple intermediate signal intensity polyps filling the nasal cavity maxillary, and ethmoid sinuses. Trapped secretions…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 708.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Noninvasive Fungal Sinusitis

IMAGING FINDINGS

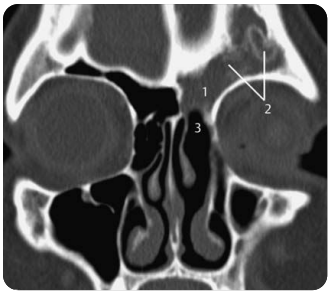
- Fungal ball: single sinus with dense contents and calcification, sclerotic walls
- Allergic fungal: expanded sinuses with central hyperdensity and low-density rim
- Very dark on T2

TOP DIFFERENTIALS

- Polyposis
- Mucocele
- Inspissated secretions

CHECKLIST

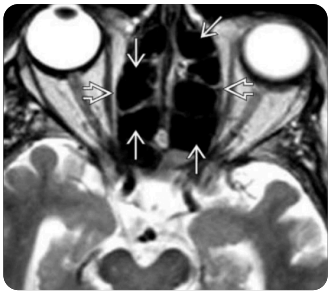
- Lamina papyracea and skull base remodelling
- Usually unilateral or asymmetric



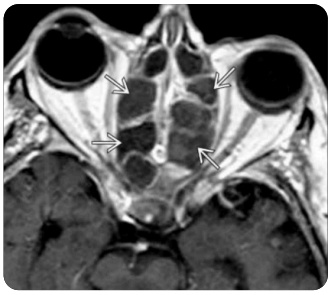
Patient with constant left frontal sinus pressure. CT, coronal (a), sagittal (b), and axial ©. There is complete opacification of the left frontal sinus…



Axial NECT shows unilateral involvement of allergic fungal sinusitis. The involved sinuses are expanded, with centrally dense inspissated…



Axial T2WI MR in a patient with bilateral AFS shows diffuse hypointense signal within the involved ethmoid air cells. There is mild sinus expansion with…



Axial T1WI C+ MR in the same patient confirms that the ethmoid sinuses are not aerated, but are in fact completely opacified. The surrounding mucosa…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 700.

NOSE & NASAL CAVITY · CONGENITAL

Choanal Atresia

IMAGING FINDINGS

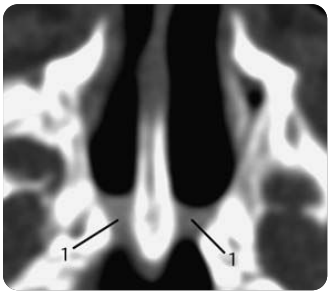
- Narrowed posterior choanae
- Thickened vomer; medially bowed posterior maxilla
- Bony, membranous or mixed closure; fluid in the nose

TOP DIFFERENTIALS

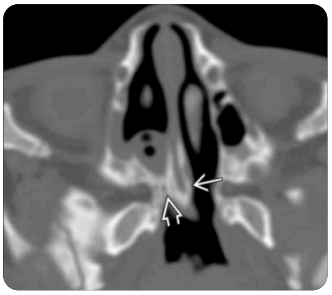
- Choanal stenosis
- Pyriform aperture stenosis
- Nasolacrimal duct mucocele

CHECKLIST

- CT after suction and decongestion
- CHARGE: look at the inner ears



CT, axial. Both choanal areas demonstrate narrowing of the bony passage, but congenital fibrous closure (1) is the cause of the distres…



Axial bone CT through the upper choana in a child with mixed bony/osseous choanal atresia shows osseous choanal obstruction…



Axial bone CT through the mid choana in the same patient shows membranous atresia bridging the narrowed inferior aspect of the…



Axial bone CT in a child with CHARGE syndrome demonstrates bilateral choanal obstruction secondary to linear membranes and mildly thickene…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 670.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Orbital Cellulitis

IMAGING FINDINGS

- Stranding of the postseptal fat
- Swollen extraocular muscles; proptosis
- No discrete collection

TOP DIFFERENTIALS

- Subperiosteal abscess
- Invasive fungal infection
- Idiopathic orbital inflammation

CHECKLIST

- Separate preseptal from postseptal
- Repeat imaging if not improving



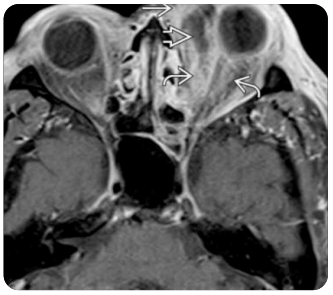
CT, axial with intravenous contrast. Bilateral opacification of the ethmoid (1). The lamina papyracea is intact without signs of periosteal…



Axial CECT in a patient with facial impetigo shows edema and enhancement of the preseptal soft tissues with normal appearance of th…



Axial STIR MR in a teenage girl shows infiltrating signal representing both preseptal and postseptal cellulitis secondary to ethmoid sinusitis.…



Axial T1-weighted post-contrast MR in a patient with recent penetrating facial trauma shows extensive preseptal and intraorbital infiltration and…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 814.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Subperiosteal Orbital Abscess

IMAGING FINDINGS

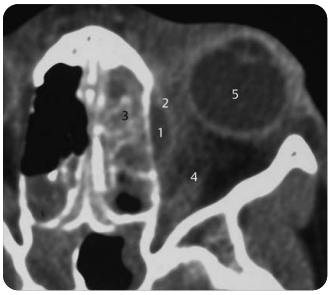
- Lentiform rim-enhancing collection along the lamina papyracea
- Medial rectus and globe displaced laterally
- Adjacent ethmoid opacification; diffusion restriction

TOP DIFFERENTIALS

- Orbital cellulitis (phlegmon)
- Idiopathic orbital inflammation
- Mucocele

CHECKLIST

- Size, posterior extent and vision
- Superior ophthalmic vein and cavernous sinus



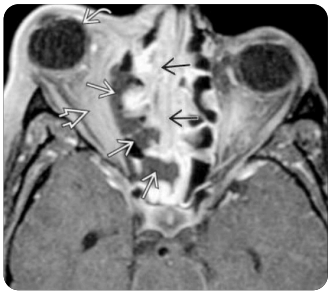
CT, axial with intravenous contrast enhancement. The CT scan shows a subperiosteal orbital abscess (1) with typical rim enhancement (2). The…



Axial CECT shows asymmetric opacification of the left ethmoid sinuses with a large subperiosteal abscess extending into the medial…



Coronal T1WI C+ FS MR in a patient status post endoscopic sinus surgery shows phlegmonous enhancement in the medial right extraconal fat with a…



Axial T1WI C+ FS MR shows a large subperiosteal abscess extending through the lamina papyracea with findings of acute sinusitis and…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 810.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Invasive Fungal Sinusitis

IMAGING FINDINGS

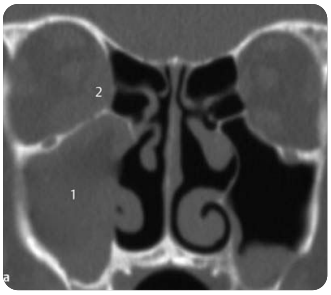
- Unilateral mucosal thickening with fat stranding outside the sinus
- Subtle bone erosion; loss of fat in the pterygopalatine fossa or orbit
- Non-enhancing mucosa ("black turbinate") on MRI

TOP DIFFERENTIALS

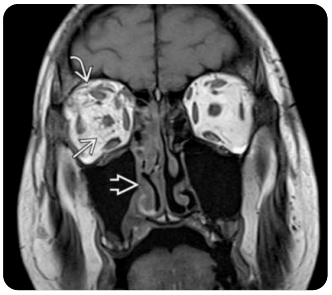
- Acute bacterial sinusitis with complication
- Lymphoma or carcinoma

CHECKLIST

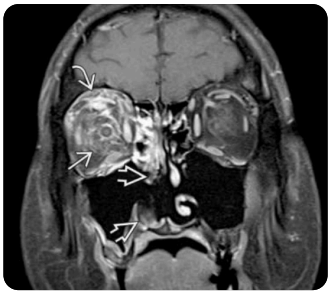
- Immunocompromised patient: endoscopy and biopsy without delay
- Orbit, cavernous sinus and brain on MRI



CT, coronal, bone-tissue setting. The bone-tissue setting reveals complete opacification of the right maxillary sinus (1), as well as very doubtful…



Coronal T1WI MR in patient with chronic myelogenous leukemia complicated by Rhizopus acute invasive fungal rhinosinusitis…



Coronal T1 C+ FS shows mucosal nonenhancement of right inferior & middle turbinates

Koch et al., Diagnostic Imaging: Head…



Axial T1WI MR in a patient with chronic myelogenous leukemia complicated by Rhizopus AIFRS shows ill-defined softtissue infiltration of the…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 704.

ORBIT & SINUS RELATIONS · INFECTION & INFLAMMATION

Mucocele

IMAGING FINDINGS

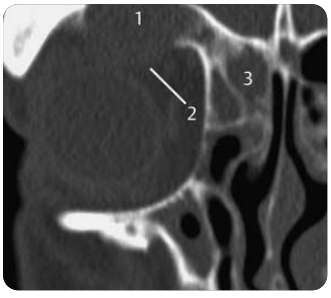
- Expanded, airless sinus with smooth remodelled walls
- Variable density and signal with protein content
- Rim enhancement only

TOP DIFFERENTIALS

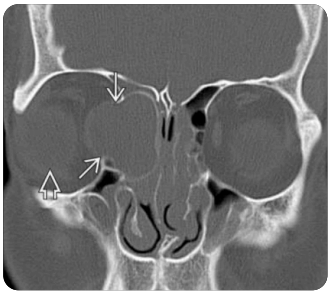
- Allergic fungal sinusitis
- Polyposis
- Tumour (solid enhancement, destruction)

CHECKLIST

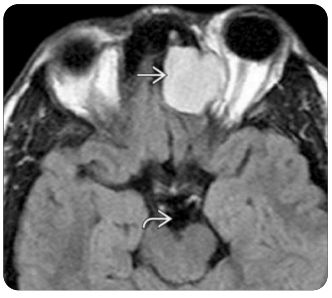
- Orbit and skull base erosion
- Frontal recess obstruction as the cause



CT, coronal. The left frontal sinus is opacified with expansion of its inferior bony outline through the roof of the orbit (1). Although this is a bone-…



Coronal bone CT shows the typical features of a right ethmoid mucocele. The sinus is opacified, and there is remodeling of the surrounding bony…



Axial FLAIR MR in a patient with proptosis and diplopia shows a homogeneous, expansile left ethmoid mucocele. Note that the material…



Sagittal T1 C+ MR demonstrates increased T1 signal in the mucocele, likely related to elevated protein content. There is no central…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 716.

ORBIT & SINUS RELATIONS · TRAUMA

Blow-out Fracture

IMAGING FINDINGS

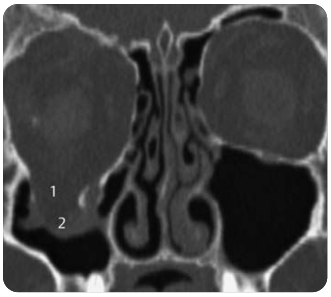
- Fracture of the orbital floor or medial wall with the rim intact
- Herniated fat and muscle; rounded inferior rectus
- Blood in the maxillary sinus; orbital emphysema

TOP DIFFERENTIALS

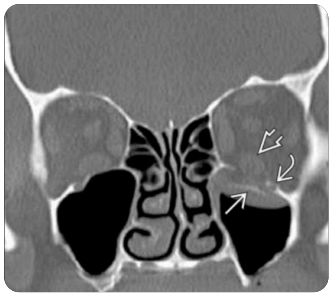
- Dehiscent lamina papyracea
- Old fracture

CHECKLIST

- Muscle entrapment, especially the trapdoor fracture in children
- Defect size (enophthalmos) and retrobulbar haemorrhage



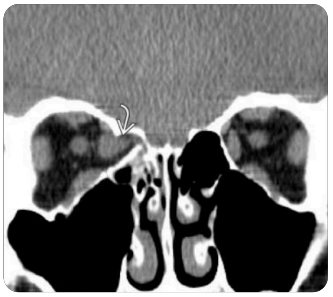
CT, coronal. Traumatic disruption of the bony floor of the orbit (1), and prolapse of orbital fat into the maxillary sinus (2). The eyeball was at a lower…



Coronal bone CT demonstrates a mildly depressed left orbital floor orbital blowout fracture
Koch et al., Diagnostic Imaging: Head…



Coronal NECT reveals a chronic right orbital floor OBF with herniation of a small volume of fat and a portion of the inferior rectus muscle through the…



Coronal soft tissue NECT shows deformity of the right medial rectus muscle, entering the osseous defect of a lamina papyracea fracture.…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 980.

NOSE & NASAL CAVITY · TUMOURS

Inverted Papilloma

IMAGING FINDINGS

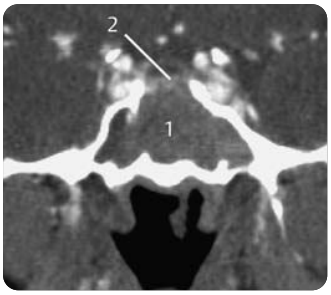
- Unilateral mass on the lateral nasal wall near the middle meatus
- Focal hyperostosis at the attachment
- Convoluted cerebriform pattern on T2 and post-contrast MRI

TOP DIFFERENTIALS

- Antrochoanal or solitary polyp
- Squamous cell carcinoma (in up to about 10%)
- Juvenile angiofibroma

CHECKLIST

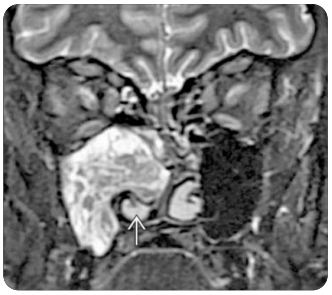
- MRI separates tumour from secretions
- Bone destruction raises concern for carcinoma



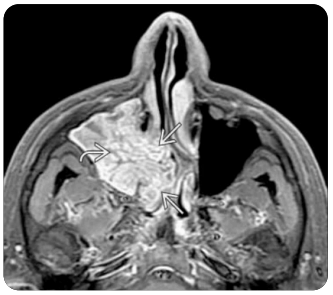
Patient operated before because of nasal polyps. Histologic evaluation revealed inverted papilloma. Follow-up scans are shown.…



Axial bone CT shows a mass in the nasal cavity along the lateral wall near the middle meatus. The maxillary sinus is opacified, but it is difficult to…



Coronal T2 FS MR shows characteristic features on an inverted papilloma that involves the right maxillary sinus and nasal cavity. The lesion has a…



Axial T1 C+ FS MR shows classic features of an inverted papilloma. The lesion is centered at the middle meatus with nasal cavity and antral…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 744.

NOSE & NASAL CAVITY · TUMOURS

Angiofibroma, Juvenile

IMAGING FINDINGS

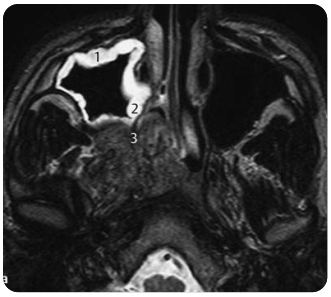
- Enhancing mass at the sphenopalatine foramen in an adolescent male
- Widened pterygopalatine fossa; bowing of the posterior maxillary wall
- Flow voids; extension to the sphenoid, infratemporal fossa and skull base

TOP DIFFERENTIALS

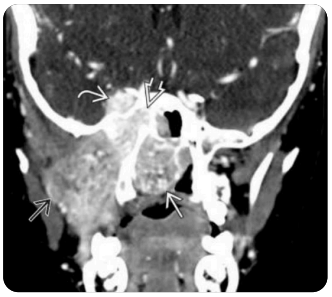
- Hypervascular polyp
- Rhabdomyosarcoma
- Antrochoanal polyp

CHECKLIST

- No biopsy: angiography and embolisation before surgery
- Vidian canal and cavernous sinus spread



A young man with obstruction and bloodtinged nasal discharge from the right side of the nose.
Dunnebier, Imaging for…



Coronal CECT in a 13-year-old shows typical growth pattern of a large, hypervascular angiofibroma occluding the nasopharynx, extending into righ…



Axial T1 C+ FS MR in a 15-year-old with nasal obstruction shows enhancing angiofibroma filling the left nasal cavity and extending into the…



Axial T1 C+ FS MR in the same patient shows a few intralesional high-flow vessels and extension of angiofibroma into the widened PPF, destruction of…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 740.

PARANASAL SINUSES · INFECTION & INFLAMMATION

Sinusitis and its orbital and intracranial complications

IMAGING FINDINGS

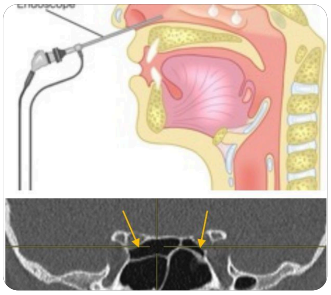
- Orbit: preseptal or postseptal stranding, subperiosteal or orbital abscess
- Intracranial: epidural or subdural empyema, abscess, meningitis
- Venous: superior ophthalmic vein and cavernous sinus thrombosis

TOP DIFFERENTIALS

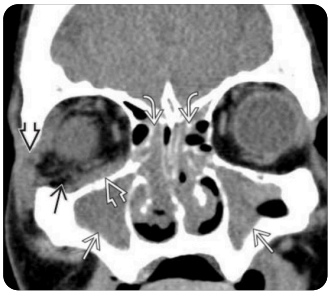
- Idiopathic orbital inflammation
- Tolosa-Hunt syndrome

CHECKLIST

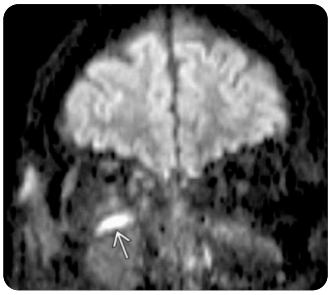
- Contrast CT for the orbit, MRI for intracranial disease
- Frontal sinusitis: look for Pott puffy tumour



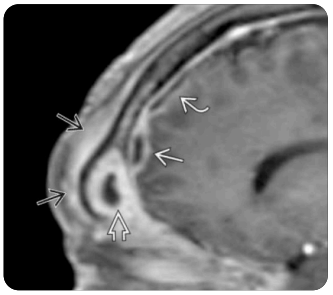
Coronal reconstructed image from a high-resolution CT. Bilateral Onodi cells (arrows). This puts the optic nerves at risk of damage during FESS…



Coronal NECT in a man with acute sinusitis complicated by orbital cellulitis and subperiosteal abscess shows extensive opacification of…



Coronal DTI trace in same patient shows reduced diffusivity, confirming subperiosteal abscess complicating acute sinusitis.…



Sagittal SPGR C+ in frontal sinusitis complicated by subdural empyema, meningitis, cerebritis, and subgaleal abscess shows frontal sinus mucosal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 696.

PARANASAL SINUSES · TUMOURS

Sinonasal tumours

IMAGING FINDINGS

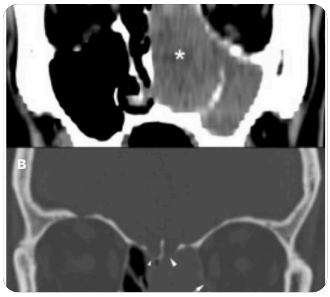
- Unilateral solid mass with bone destruction
- Intermediate T2, solid enhancement, distinct from bright secretions
- Spread to orbit, skull base, pterygopalatine fossa, cheek

TOP DIFFERENTIALS

- Inverted papilloma
- Lymphoma
- Olfactory neuroblastoma, adenocarcinoma, melanoma

CHECKLIST

- Periorbita and dura
- Perineural spread along V2 and the vidian nerve; nodes



Coronal reformatted images from a noncontrast CT in soft tissue window (a) and bone window settings (b). The mass (asterisk) causes resorption of…



Axial CECT shows the typical location and appearance of an antral SCCa. There is extension into the premaxillary soft tissues anteriorly an…



Axial T1 MR shows a large antral SCCa. The signal of the mass is similar to other soft tissues. There is extension anteriorly into the premaxillary soft…



Coronal T2 FS MR of SCCa demonstrates ethmoid sinus involvement and masticator space extension. The low T2 signal of this…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 754.

IMAGING

Advanced and rare

Once the core is secure. Each has its full set of images in the app.

Meningioma with Skull Base Destruction

Paranasal sinuses · Tumours

On CT, the bone changes associated with hyperostosing meningioma en plaque may be confused with (polyostotic) fibrous dysplasia. However, in meningioma the cortical margins of the affected bones are not well delineated. · In fibrous dysplasia, the cortical margins are preserved (compare with section on fibrous dysplasia). Cranial nerves and intracranial vasculature may be encroached on by the lesion. Total surgical removal is rarely possible.

Mucocele of the Frontal Recess

Paranasal sinuses · Infection & inflammation

Beware of large osseous lesions of the skull base, removal of which may result in protrusion of intracranial contents or marsupialization of the mucocele. · In meningo(encephalo)cele or intracranial tumors, persistent leakage of cerebrospinal fluid may occur at biopsy.

Olfactory Neuroblastoma (Olfactory neuroblastoma (esthesioneuroblastoma))

Nose & nasal cavity · Tumours

Olfactory neuroblastoma (esthesioneuroblastoma) is an uncommon tumor of neural crest origin that arises from the olfactory mucosa in the superior nasal cavity. These polypoid tumors typically start unilaterally and may bleed profusely on biopsy. Neglected cases appear as a bilateral mass in the anterior cranial fossa. In these larger tumors, which usually have intracranial extension, peripheral tumoral cysts can occur at the margins of the intracranial part of the mass. These cysts have their broadest base on the tumor and, when present, are highly suggestive of the (imaging) diagnosis of olfactory neuroblastoma. Beware of encroachment of cranial nerves or intracranial vessels, which may be important in the estimation of morbidity during removal.

For personal study. Follow local protocols and senior advice for patients.

Pituitary Gland Pathology

Paranasal sinuses · Tumours

In classifications of pituitary gland pathology, lesions within the sella are classified as microadenomas. Microadenomas are frequently active hormone-producing tumors with specific clinical symptoms (e.g., galactorrhea). · Tumors extending out of the sella are classified as macroadenomas. Most macroadenomas are nonfunctional tumors. Therefore, these lesions present late and can reach remarkable volumes. Large pituitary adenomas may be accompanied by cystic degeneration with or without hemorrhage. Macroadenomas with superior extension may compromise the optic chiasm with visual field disturbances and even blindness. Less frequently, macroadenomas spread inferiorly into the sphenoid sinus.

Intracranial Abscess from a Sinonasal Focus

Paranasal sinuses · Infection & inflammation

Frontal and ethmoid sinusitis spread through valveless diploic veins or bone erosion to cause epidural, subdural or frontal lobe abscess (Pott's puffy tumour is the extracranial sign). · Contrast MRI is more sensitive than CT for subdural empyema and cerebritis; repeat imaging if seizures or reduced consciousness develop after sinus surgery.

Dural Defect after Sinonasal Surgery

Paranasal sinuses · After surgery

The commonest site of iatrogenic skull base injury is the lateral lamella of the cribriform plate (thinnest bone, where the anterior ethmoidal artery enters), then the posterior ethmoid roof. · Locate the defect with thin-section coronal CT; MR cisternography (heavily T2-weighted) shows CSF tracking through it; beta-2 transferrin confirms leaked fluid.

Fracture of the Nasal Bone and Retention Cysts

Nose & nasal cavity · Trauma

In cases of large cysts resulting in obstruction of the maxillary sinus and those accompanied by signs of sinusitis, surgical removal (marsupialization) might be considered to improve drainage.

Unilateral Nasal Polyps

Nose & nasal cavity · Infection & inflammation

Destruction of bony outlines (with infiltrative characteristics) is suggestive of inverted papilloma, malignancies, or fungal infection. · Calcifications in the opacified area are suggestive of fungal infection. Resorption of bone around tooth roots suggests odontogenic infection.

Frontal Sinusitis

Paranasal sinuses · Infection & inflammation

The anatomy and pathology of the frontal recess required special attention. · In cases of persisting drainage problems in the region of the frontal recess, the septum between the frontal sinuses may be removed for establishing drainage by way of the left frontal recess.

Odontogenic Sinusitis

Paranasal sinuses · Infection & inflammation

Destruction of bony outlines with infiltrative characteristics is suggestive of inverted papilloma, malignancies, or fungal infections. · Calcifications in the opacified area are suggestive of fungal infections.

Cystic Fibrosis

Paranasal sinuses · Infection & inflammation

Destruction of bony outlines with infiltrative signs is more characteristic of malignancy. · Expansion is more likely to be due to benign processes, such as polyposis or mucocele.

Granulomatosis with Polyangiitis (Wegener)

Nose & nasal cavity · Infection & inflammation

A history of chronic sinusitis in combination with destruction of the nasal septum is highly suggestive of granulomatosis with polyangiitis (GPA). Pulmonary complaints and rapid improvement after steroid treatment may confirm this diagnosis. These therapeutic effects may also be observed in cases of allergic fungal sinusitis. · Orbital involvement appears as a mass resulting in protrusion of the eyeball and diplopia (see also granulomatosis with polyangiitis (GPA), Orbital Involvement).

Fibrous Dysplasia of the Sinuses (Monostotic and Polyostotic)

Paranasal sinuses · Bone disease

· Expansion of bone with a ground glass appearance and intact cortex reflects the pathophysiology of this disease (i.e., replacement of normal medullary bone by woven fibro-osseous tissue).

Osteoma

Paranasal sinuses · Bone disease

Large osteomas may cause functional problems due to growth and encroachment of surrounding structures (see Osteoma, Gardner Syndrome, and Complications). For this reason, and also in small osteomas, a wait-and-scan policy might be advisable.

Piriform Aperture Malformation

Nose & nasal cavity · Congenital

A developmental disorder of this kind may result in concomitant pathologies, such as the fusion of both maxillary central incisors to form one megaincisor (Fig. 7.15), in the majority of patients. Holoprosencephaly and abnormalities of the pituitary–adrenal axis are also encountered.

Nasolacrimal Mucocele

Orbit & sinus relations · Congenital

Myxoma

Paranasal sinuses · Tumours

Osteoma, Gardner Syndrome, and Complications

Orbit & sinus relations · Bone disease

Extension to structures which, at surgical removal, may result in cosmetic or functional complications. · In Gardner syndrome, multiple osteomas may be found as well as other pathologies, some of which are at an increased risk of malignancy (colorectal carcinoma and thyroid carcinoma).

Granulomatosis with Polyangiitis, Orbital Involvement

Nose & nasal cavity · Infection & inflammation

Amyloidosis in the Orbit

Orbit & sinus relations · Infection & inflammation

Differentiation between solitary involvement of the orbit and involvement or growth from adjacent structures. · Differentiation between unilateral and bilateral disease.

Anterior Skull Fractures

Orbit & sinus relations · Trauma

Encroachment of functional structures (ocular muscles, cranial nerves) or compression due to hematoma (orbit, subdural). · Displacements resulting in functional (movements of the eyeball, obstruction of the nose) or cosmetic deficits.

Orbital Decompression in Graves Disease

Orbit & sinus relations · After surgery

Conventional sinonasal radiography in these patients is not sufficient for the evaluation of concomitant sinusitis. CT will be needed for anatomic orientation, especially if endoscopic surgery is contemplated. · Endonasal surgical procedures must be done with care due to distortion of the normal anatomy and an increased risk of damage to displaced intraorbital structures such as the optic nerve.